

EBC4097 — Supplementary Study Resources

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How to use these resources

Use these to build intuition after each chapter of the study book; for the exam, the lecture slides and the study book remain the reference.

Outside sources often use different symbols (for example ϕ instead of α_1 , or swapped letters in GARCH). When notation differs, follow the slides.

1. Read a chapter of the study book.
2. Watch the matching short video (ritvikmath or StatQuest), about 10–15 minutes per topic.
3. Open a textbook (FPP or ISLR) only for topics that still feel shaky.
4. Keep the MIT lectures and Engle's Nobel lecture for depth, for example during the group assignment.

Quick map: chapter to resource

Each study-book chapter has one video to watch and one reading to open if needed.

Study-book chapter	Topic	Watch	Read
1–3	Time series, stationarity, ACF/PACF	ritvikmath: Stationarity; White Noise; Autocorrelation and Partial Autocorrelation	FPP2 Ch. 8.1
4	AR, MA, ARMA, forecasting	ritvikmath: Autoregressive Model	FPP2 Ch. 8.1–8.7
5	Trends, unit roots, Dickey-Fuller	ritvikmath: Unit Roots; Augmented Dickey Fuller Test + Code	FPP2 Ch. 8.1

Study-book chapter	Topic	Watch	Read
7	VAR, Granger causality, impulse responses	ritvikmath: Vector Auto Regression; MIT Time Series II	FPP2 Ch. 11.2; vars article
8	ARCH, GARCH, risk	ritvikmath: ARCH Model; GARCH Model; MIT Volatility Modeling	Engle's Nobel lecture
9	Ridge, Lasso, PCA	StatQuest: Ridge, Lasso, Ridge vs Lasso, PCA Step-by-Step	ISLR chapters on regularization and unsupervised learning
App. A	R programming	StatQuest R videos; ISLR labs	Smeekes R training

From your course

These follow the course most closely, because your syllabus and tutorial sheets cite them directly.

- [Stephan Smeekes' R training](#): sessions 1–4 prepare the R basics tutorial, session 5 the intermediate one.
- [Forecasting: Principles and Practice, 2nd edition](#): the ARMA tutorial cites chapters 8.1–8.7; the VAR tutorial cites chapter 11.2 ([ARIMA in R](#)).
- r-econometrics.com, cited in the VAR tutorial: [introduction to VAR models](#) and [impulse responses](#).

Short videos for intuition: ritvikmath

ritvikmath's "Time Series Talk" series on YouTube is the best beginner fit; its videos on the autoregressive model, white noise and VAR rank among the [most-watched time series explainers](#). Search YouTube for "Time Series Talk" plus the topic:

- Stationarity; White Noise; Autocorrelation and Partial Autocorrelation (Chapters 2–3)
- Autoregressive Model (Chapter 4)
- Unit Roots; Augmented Dickey Fuller Test + Code (Chapter 5)

- Vector Auto Regression (Chapter 7)
- ARCH Model; GARCH Model; Coding the GARCH Model (Chapter 8)

The coding videos use Python, not R; watch them for the ideas, and use your course R scripts for practice.

Free textbook: Forecasting: Principles and Practice

Use the 2nd edition: it matches the forecast package your tutorials use.

Hyndman and Athanasopoulos' book is a comprehensive, free online introduction to forecasting with R examples ([3rd edition](#)). The [2nd edition](#) uses the forecast package; the 3rd uses tsibble, feasts and fable ([OTexts](#)). Best for Chapters 4–5 of the study book: stationarity, differencing, ARMA models and forecast evaluation.

University lectures: MIT 18.S096

MIT's Topics in Mathematics with Applications in Finance (Fall 2013, Dr. Peter Kempthorne) has free video lectures that go deeper than the course; watch them after the study book.

Lecture	Covers	Study-book chapter
Time Series Analysis I	Stochastic processes, regression and stationarity models	2–4
Time Series Analysis II	Multivariate time series, least-squares estimation	7
Time Series Analysis III	Cointegration, cointegrated VAR models	5, 7
Volatility Modeling	Historical volatility and volatility models	8

The [full YouTube playlist](#) also has lectures on Value at Risk and factor modeling (Chapters 8–9).

Volatility from the inventor: Engle's Nobel lecture

Robert Engle won the 2003 Nobel Prize for ARCH; his free Prize Lecture, [Risk and Volatility: Econometric Models and Financial Practice](#), is the best background reading for Chapter 8.

Engle once [summed up his work for reporters](#): when volatility is low, it will probably stay low; when it is high, it will probably stay high for a while before coming down. That is volatility clustering in one sentence.

Ridge, Lasso and PCA: ISLR and StatQuest

For Chapter 9, start with StatQuest's short videos, then use ISLR for depth.

StatQuest (Josh Starmer, YouTube; [video list](#)), in this order:

1. Regularization Part 1: Ridge (L2) Regression
2. Regularization Part 2: Lasso (L1) Regression
3. Ridge vs Lasso Regression, Visualized!!! (pairs with the diamond-and-circle figure in the book)
4. Ridge, Lasso and Elastic-Net Regression in R
5. PCA main ideas in only 5 minutes!!!
6. Principal Component Analysis (PCA), Step-by-Step

[An Introduction to Statistical Learning](#) (James, Witten, Hastie, Tibshirani): free PDF, with chapters on linear model selection and regularization and on unsupervised learning, each with an R lab. The authors' [video lectures](#) include Shrinkage Methods and Ridge Regression, The Lasso, Tuning Parameter Selection for Ridge Regression and Lasso, and Unsupervised Learning and Principal Components Analysis.

VAR in R: the vars package

For Chapter 7 and the group assignment, Bernhard Pfaff's free article [VAR, SVAR and SVEC Models: Implementation Within R Package vars](#) is the reference for the package your tutorials use.

It covers estimation, diagnostic testing, prediction, Granger causality, impulse responses and forecast error variance decomposition. The package itself is on [CRAN](#).